



St Aloysius' College
Year 12 Mid-Year Examinations
2009

EXTENSION 2 MATHEMATICS
(Additional paper)

Total marks: 75

Reading time - 5 minutes

Working time – 2 hours

Examination papers must NOT be removed
from the examination room.

General Instructions

- Start each question in a new booklet
- Board approved calculators may be used
- Marks may be deducted for careless or badly arranged work
- Show all necessary work

SUPERVISOR'S INSTRUCTIONS:

Please issue five 4 page answer booklets.

Please collect the examination paper with the answer booklets.

Question 1 (15 marks)

(a) If $z = 2 - i$,

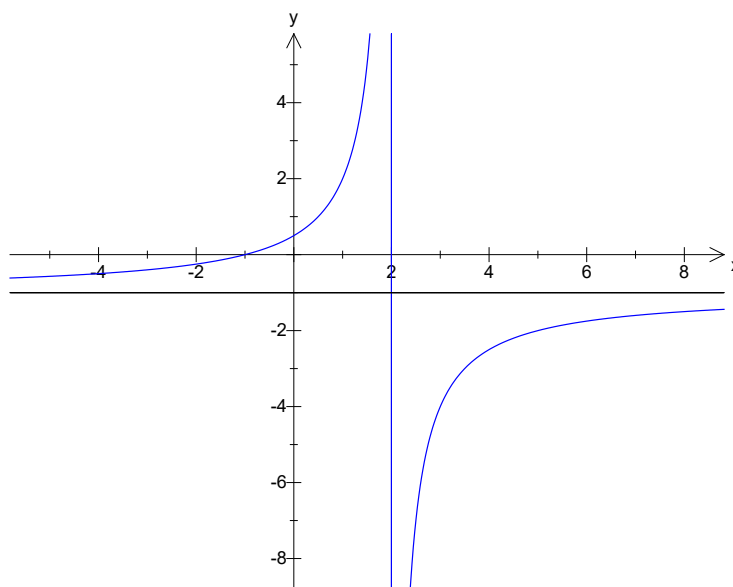
(i) Find $\frac{1}{z}$ in the form $a + ib$. 1

(ii) Find the real numbers p and q such that $pz + \frac{q}{z} = 1$. 2

(b) Use De Moivre's theorem to evaluate $(1 + i)^7 + (1 - i)^7$. 3

(c) If $|z - 2 - i| = 2$, find the maximum and minimum values of $|z|$. 3

(d) The graph for $y = \frac{x+1}{2-x}$ is shown below.



On the separate answer sheet provided sketch the graphs of the following:

(i) $y^2 = \frac{x+1}{2-x}$ 2

(ii) $|y| = \frac{x+1}{2-x}$ 2

(iii) $|y| = \frac{|x|+1}{2-|x|}$ 2

Question 2 (15 marks) START A NEW BOOKLET

(a) Sketch the locus of z for each of the following:

(i) $2(z + 2) - 5i(z - 2) = 20$ 2

(ii) $\frac{\pi}{4} \leq \arg\left(\frac{1}{z}\right) \leq \frac{\pi}{2}$ 3

(b) If $1, \omega, \omega^2$ are the three cube roots of 1, evaluate

(i) $(1 + \omega)^9$ 3

(ii) $\omega^2 + \omega + 1 + \frac{1}{\omega} + \frac{1}{\omega^2}$ 2

(c) Use De Moivre's Theorem to solve $z^4 = -4i$, showing all steps in your reasoning. Write your answers in mod-arg form and show the solutions clearly on an Argand diagram. 5

Question 3 (15 marks) START A NEW BOOKLET

(a) Find the equation of the ellipse with the point $(1, 2)$ as centre, the line $4x = 29$ as directrix, and eccentricity $\frac{4}{5}$. 3

(b) For the curve $y = \frac{x^3 - 4}{x^2}$

(i) Find any stationary points and intercepts. 3

(ii) Find the equations of any asymptotes for the curve $y = \frac{x^3 - 4}{x^2}$ 2

(iii) Sketch the curve $y = \frac{x^3 - 4}{x^2}$. 2

(iv) From the graph find the values of k for which the equation $x^3 - kx^2 - 4 = 0$ has three distinct real roots. 2

(c) α, β, γ are three complex numbers represented by the points A, B, C on the Argand diagram and $\frac{\alpha - \gamma}{\beta - \gamma} = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$.

Show that the triangle ABC is equilateral. 3

Question 4 (15 marks) START A NEW BOOKLET

- (a) (i) Find an expression for the derivative for the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. 2
- (ii) Hence show that, at the point in the first quadrant where $x = \frac{a}{2}$, the gradient of the tangent to the ellipse is $-\frac{b}{a\sqrt{3}}$. 2
- (b) For the hyperbola $4x^2 - y^2 = 100$ find the
- (i) eccentricity, 3
- (ii) coordinates of the foci, 1
- (iii) equations of the directrices, 1
- (iv) equations of the asymptotes. 1
- (c) The locus defined by the equation $|z - 1| - |z + 3| = 2$ in the complex number plane gives a conic.
Find the cartesian equation of the conic and state any restrictions. 5

Question 5 is on the next page

Question 5 (15 marks) START A NEW BOOKLET

- (a) θ is real and constant and $0 \leq \theta < \frac{\pi}{2}$. Solve the following equation for z over the complex numbers. Give your answer in simplest form.

$$z^2 \cos^2 \theta + z \sin 2\theta + 1 = 0 \quad 3$$

- (b) For the curve $\ln y = \cos^{-1} x$

(i) State the domain and range, 3

(ii) Find any intercepts, stationary points or vertical tangents, 3

(iii) Sketch the curve showing the above features. 2

- (c) A particle is projected from a point at ground level with speed V and making an angle of α with the horizontal. After time t its horizontal and vertical displacements are x and y respectively. As a consequence the cartesian equation of the projectile's path is

$$y = x \tan \alpha - \frac{g \sec^2 \alpha}{2V^2} x^2.$$

The projectile passes through the point $P(a, h)$.

Prove that there are two different paths for any given initial speed V if

$$(V^2 - gh)^2 > g^2(a^2 + h^2). \quad 4$$

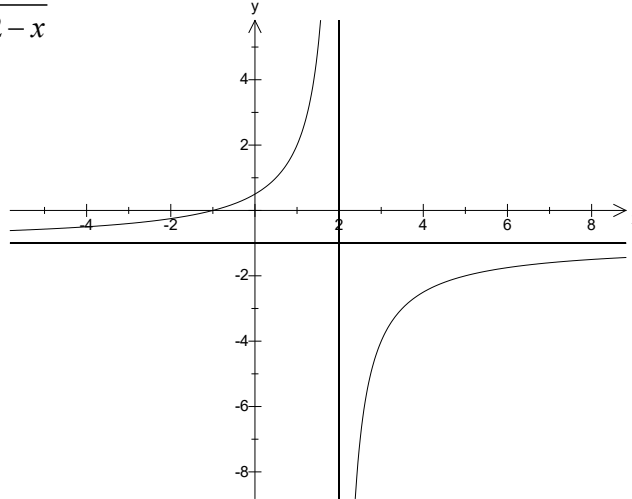
END OF EXAM

Student Number:

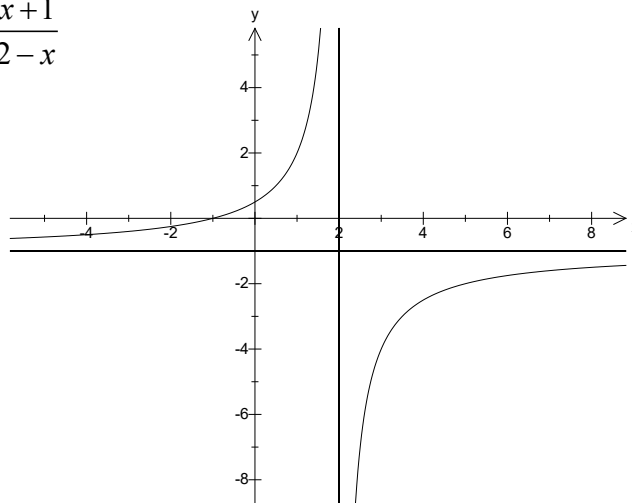
Hand in this page with your answer booklets.

Question 1 (d) Use the graphs of $y = \frac{x+1}{2-x}$ as a template to sketch each of the following:

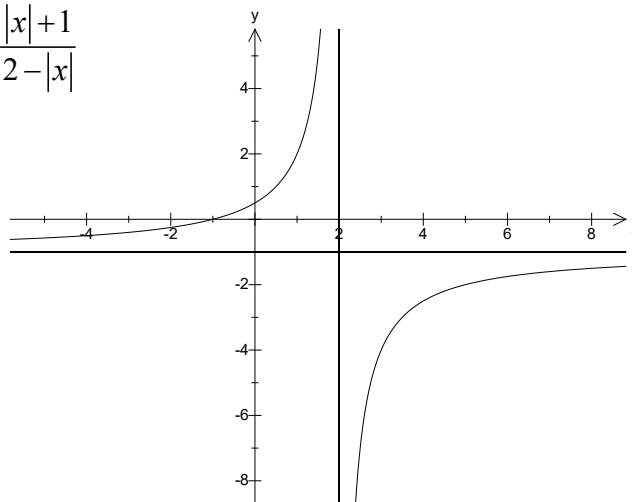
(i) $y^2 = \frac{x+1}{2-x}$



(ii) $|y| = \frac{x+1}{2-x}$



(iii) $|y| = \frac{|x|+1}{2-|x|}$



$$\begin{aligned}
 \text{Q1 a) i)} \quad \frac{1}{2-i} &= \frac{2+i}{(2-i)(2+i)} \\
 &= \frac{2+i}{2^2 - i^2} \\
 &= \frac{2+i}{5} \\
 &= \frac{2}{5} + \frac{i}{5} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad \frac{pz+q}{z} &= p(2-i) + \frac{q}{2-i} \\
 &= 2p - pi + q\left(\frac{2}{5} + \frac{i}{5}\right) \\
 &= 2p + \frac{2q}{5} - pi + \frac{qi}{5} \quad \checkmark
 \end{aligned}$$

$$\text{but } \frac{pz+q}{z} = 1 + 0i$$

$$\therefore 2p + \frac{2q}{5} = 1 \quad \text{and} \quad -p + \frac{q}{5} = 0$$

$$\therefore p = \frac{q}{5}$$

$$\frac{2q}{5} + \frac{2q}{5} = 1$$

$$\frac{4q}{5} = 1$$

$$q = \frac{5}{4}$$

$$p = \frac{1}{4} \quad \checkmark$$

Q 1 (b) $(1+i) = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$

$(1-i) = \sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4} \right)$ ✓

$(1+i)^7 + (1-i)^7 = (\sqrt{2})^7 \operatorname{cis} \frac{7\pi}{4} + (\sqrt{2})^7 \operatorname{cis} \left(-\frac{7\pi}{4} \right)$ ✓

$= 8\sqrt{2} \left[\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} + \cos \left(-\frac{7\pi}{4} \right) + i \sin \left(-\frac{7\pi}{4} \right) \right]$

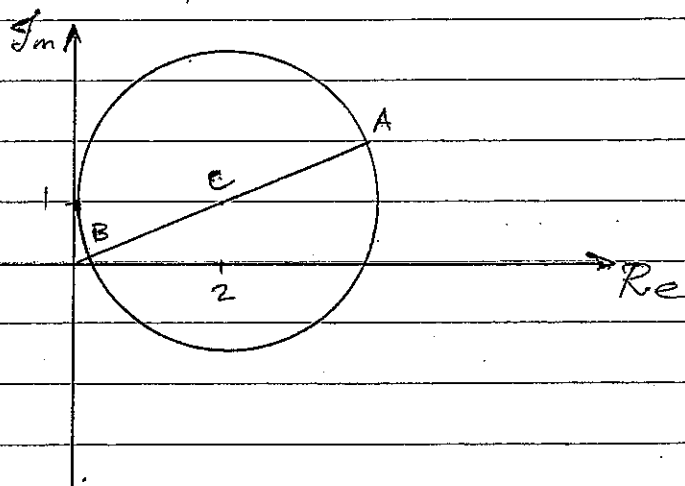
$= 8\sqrt{2} \left[\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} + \cos \frac{7\pi}{4} - i \sin \frac{7\pi}{4} \right]$

$= 8\sqrt{2} \times 2 \cos \frac{7\pi}{4}$

$= 8\sqrt{2} \times 2 \times \frac{1}{\sqrt{2}}$

$= 16$ ✓

(c) $|z - (2+i)| = 2$ gives a circle centre $(2,1)$ $r = 2$



Distance $OC = \sqrt{2^2 + 1^2}$
 $= \sqrt{5}$

∴ At A $\max |z| = \sqrt{5} + 2$ ✓

At B $\min |z| = \sqrt{5} - 2$ ✓

Q 2 (a) (i)

$$\text{If } z = x + iy$$

$$\bar{z} = x - iy$$

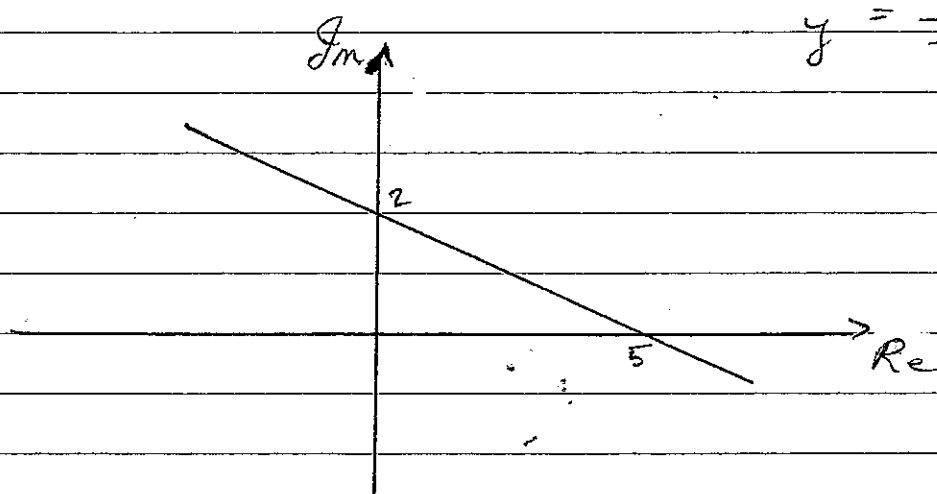
$$z + \bar{z} = 2x$$

$$z - \bar{z} = 2iy$$

$$\therefore 2(z + \bar{z}) - 5i(z - \bar{z}) = 4x - 5i(2iy) = 4x + 10y = 20$$

$$2x + 5y = 10$$

$$y = \frac{-2x + 20}{5}$$



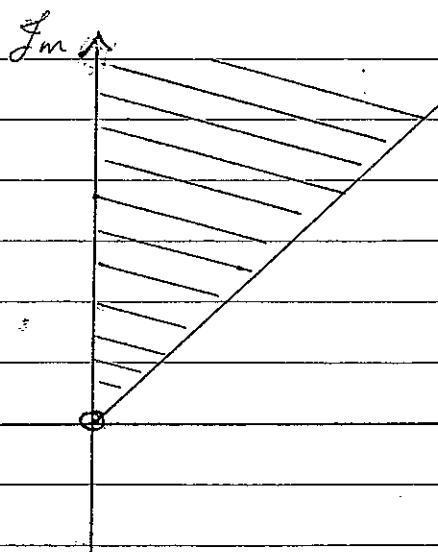
ii)

$$\arg\left(\frac{1}{\bar{z}}\right) = \arg 1 - \arg \bar{z}$$

$$= 0 - \arg \bar{z}$$

$$= \arg z$$

$$\therefore \frac{\pi}{4} \leq \arg z \leq \frac{\pi}{2}$$



Note $z \neq 0$.

$$b) \quad i) \quad 1 + \omega + \omega^2 = 0$$

$$\begin{aligned} 1 + \omega &= -\omega^2 \\ (1 + \omega)^9 &= (-\omega^2)^9 \\ &= -\omega^{18} \\ &= -(\omega^3)^6 \\ &= -1^6 \\ &= -1 \end{aligned}$$

$$\begin{aligned} ii) \quad \omega^2 + \omega + 1 + \frac{1}{\omega} + \frac{1}{\omega^2} &= 0 + \frac{1}{\omega} + \frac{1}{\omega^2} \\ &= 0 + \frac{\omega^2}{\omega^3} + \frac{\omega}{\omega^3} \\ &= 0 + \frac{\omega^2 + \omega}{1} \\ &= \omega^2 + \omega \\ &= -1 \end{aligned}$$

$$\begin{aligned} c) \quad |z^4| &= |-4i| \\ |z|^4 &= 4 \\ |z| &= \sqrt[4]{4} \\ &= \sqrt{2} \quad \text{since } |z| \geq 0. \end{aligned}$$

$$\begin{aligned} \text{Then } z &= \sqrt{2} \operatorname{cis} \theta \\ z^4 &= 4 \operatorname{cis} 4\theta \\ &= 4 \cos 4\theta + 4i \sin 4\theta = 0 + -4i \end{aligned}$$

$$\text{Then } \cos 4\theta = 0 \quad \text{and} \quad \sin 4\theta = -1$$

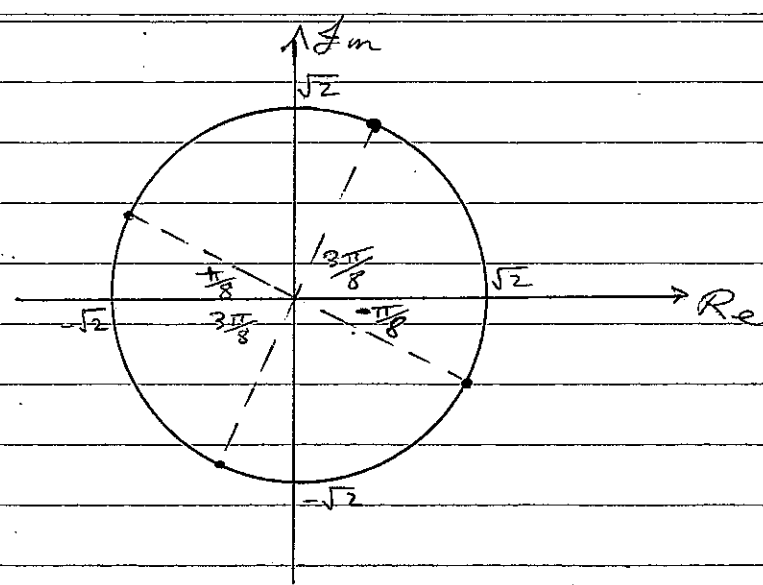
$$4\theta = \frac{-\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{2}, \frac{11\pi}{2}$$

$$\theta = \frac{-\pi}{8}, \frac{3\pi}{8}, \frac{7\pi}{8}, \frac{11\pi}{8}$$

$$\text{OR } \theta = \frac{-5\pi}{8}, \frac{-\pi}{8}, \frac{3\pi}{8}, \frac{7\pi}{8}$$

$$z = \sqrt{2} \operatorname{cis} \left(\frac{-5\pi}{8} \right), \sqrt{2} \operatorname{cis} \left(\frac{-\pi}{8} \right), \sqrt{2} \operatorname{cis} \frac{3\pi}{8}, \sqrt{2} \operatorname{cis} \frac{7\pi}{8}$$

Q2 c)



Q 3 (a) Directrix $x = \frac{29}{4}$

$$1 + \frac{a}{e} = \frac{29}{4}$$

$$\frac{a}{e} = \frac{25}{4}$$

but $e = \frac{4}{5}$

$$\therefore \frac{5a}{4} = \frac{25}{4}$$

$$a = 5$$

$$b^2 = a^2(1 - e^2)$$

$$= 25 \left(1 - \frac{16}{25}\right)$$

$$= 9$$

$$b = 3$$

Ellipse $\frac{(x-1)^2}{5^2} + \frac{(y-2)^2}{3^2} = 1$

$$\frac{(x-1)^2}{25} + \frac{(y-2)^2}{9} = 1$$

(b) i)

$$y = x - \frac{4}{x^2}$$

$$\frac{dy}{dx} = 1 + 8x^{-3}$$

$$= 1 + \frac{8}{x^3} = 0$$

$$x^3 = -8$$

$$x = -2$$

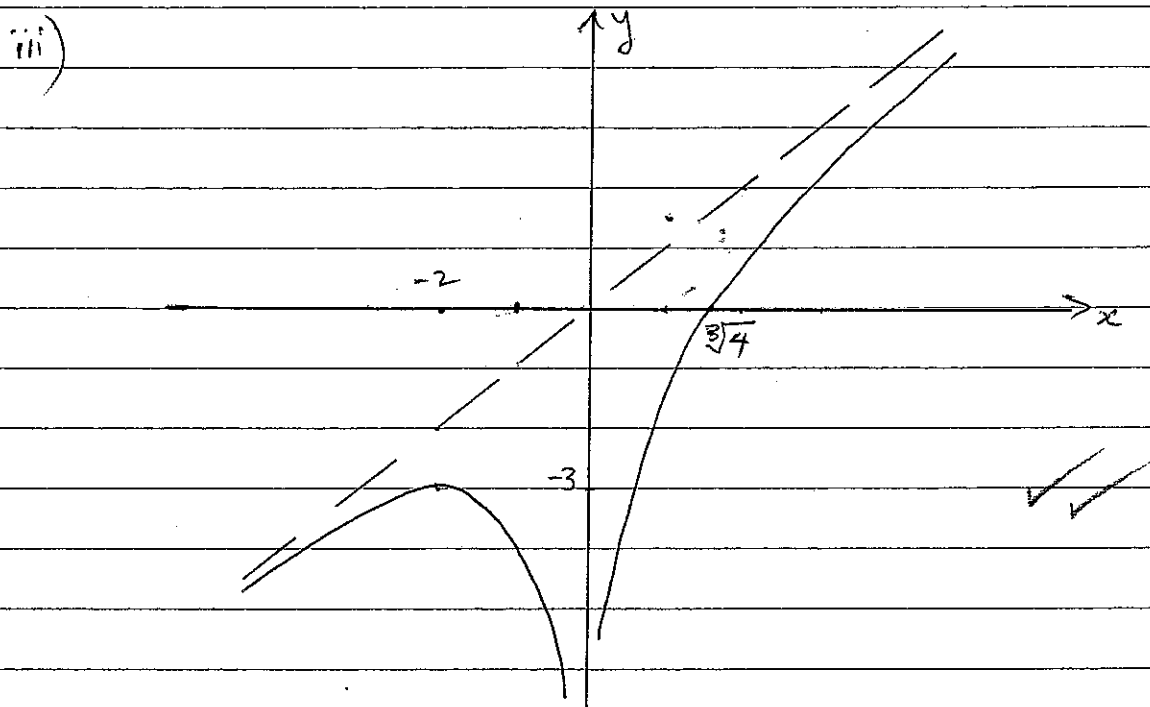
$$y = -3$$

Stat point $(-2, -3)$

(i) $x = 0$ y undef. ✓
 $y = 0$ $x = \sqrt[3]{4} \approx 1.59$ ✓

ii) $y = x - \frac{4}{x^2}$
 $y \rightarrow x$ as $x \rightarrow \pm \infty$.

Asymptotes: $y = x$ ✓
 $x = 0$. ✓



iv) $x^3 - 4 = kx^2$
 $\frac{x^3 - 4}{x^2} = k$

$y = k$ ✓

$k < -3$ gives 3 intersection points ✓

Q 3 c) $\arg \frac{\alpha - \gamma}{\beta - \gamma} = \frac{\pi}{3}$

$$\arg(\alpha - \gamma) - \arg(\beta - \gamma) = \frac{\pi}{3}$$

$$\arg(\alpha - \gamma) = \arg(\beta - \gamma) + \frac{\pi}{3}$$

$$\therefore \angle ACB = \frac{\pi}{3}$$

Also $\left| \frac{\alpha - \gamma}{\beta - \gamma} \right| = \left| \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right|$

$$\frac{|\alpha - \gamma|}{|\beta - \gamma|} = 1$$

$$|\alpha - \gamma| = |\beta - \gamma|$$

$$CA = CB$$

$\therefore \triangle ACB$ is isosceles with $\angle ACB = 60^\circ$

$\therefore \triangle ACB$ is equilateral.

Q4 (a) (i) $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0.$

$$\frac{y}{b^2} \frac{dy}{dx} = -\frac{x}{a^2}$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

ii) $x = \frac{a}{2}$ $\frac{y^2}{b^2} = 1 - \left(\frac{a}{2}\right)^2 \frac{1}{a^2}$

$$= 1 - \frac{1}{4}$$

$$= \frac{3}{4}$$

$$y^2 = \frac{3b^2}{4}$$

$$y = \pm \frac{b\sqrt{3}}{2}$$

but in 1st quadrant point is $\left(\frac{a}{2}, \frac{b\sqrt{3}}{2}\right)$

then $m = \frac{-b^2}{a^2} \times \frac{a}{2} \times \frac{2}{b\sqrt{3}}$

$$= \frac{-b}{a\sqrt{3}}$$

~~(b) (i) $\frac{y^2}{100} - \frac{x^2}{25} = 1.$~~

Q 4 (b) i) $4x^2 - y^2 = 100$
 $\frac{x^2}{25} - \frac{y^2}{100} = 1.$

$a = 5, \quad b = 10$

$b^2 = a^2(e^2 - 1)$

$100 = 25(e^2 - 1)$

$4 = e^2 - 1$

$e^2 = 5$

$e = \sqrt{5}.$

ii) foci $(\pm 5\sqrt{5}, 0)$

iii) directrices $x = \pm \frac{5}{\sqrt{5}}$ or $x = \pm \sqrt{5}.$

iv) asymptotes $y = \pm \frac{b}{a}x$
 $= \pm 2x.$

c) Hyperbola with foci $(1, 0)$ $(-3, 0)$
 Centre $(-1, 0)$

$2a = 2$

$a = 1$

$ae = 2$

$e = 2.$

$b^2 = a^2(e^2 - 1)$

$b^2 = 1(4 - 1)$
 $= 3$

$b = \sqrt{3}.$

Equation $\frac{(x+1)^2}{1^2} - \frac{y^2}{(\sqrt{3})^2} = 1$

$\frac{(x+1)^2}{3} - \frac{y^2}{3} = 1.$

Locus gives left/negative branch only

c) ALT

$$\text{If } z = x + iy, \quad \sqrt{(x-1)^2 + y^2} - \sqrt{(x+3)^2 + y^2} = 2. \quad \checkmark$$

$$\begin{aligned} \sqrt{(x-1)^2 + y^2} &= 2 + \sqrt{(x+3)^2 + y^2} \\ (x-1)^2 + y^2 &= 4 + 4\sqrt{(x+3)^2 + y^2} + (x+3)^2 + y^2 \\ x^2 - 2x + 1 + y^2 &= x^2 + 6x + 9 + y^2 + 4 + 4\sqrt{(x+3)^2 + y^2} \\ -8x - 12 &= 4\sqrt{(x+3)^2 + y^2} \quad \checkmark \\ -2x - 3 &= \sqrt{(x+3)^2 + y^2} \\ 4x^2 + 12x + 9 &= (x+3)^2 + y^2 \\ &= x^2 + 6x + 9 + y^2 \quad \checkmark \end{aligned}$$

$$3x^2 + 6x - y^2 = 0.$$

$$x^2 + 2x - \frac{y^2}{3} = 0$$

$$x^2 + 2x + 1 - \frac{y^2}{3} = 1$$

$$(x+1)^2 - \frac{y^2}{3} = 1. \quad \checkmark$$

Restriction

Q5 a)

$$z^2 \cos^2 \theta + 2 \sin \theta \cos \theta z + 1 = 0$$

$$z^2 \cos^2 \theta + 2 \sin \theta \cos \theta z + \sin^2 \theta = \sin^2 \theta - 1$$

$$(z \cos \theta + \sin \theta)^2 = -(1 - \sin^2 \theta)$$

$$= -\cos^2 \theta$$

$$z \cos \theta + \sin \theta = \pm i \cos \theta$$

$$z \cos \theta = -\sin \theta \pm i \cos \theta$$

$$z = -\tan \theta \pm i$$

ALT

$$z = \frac{-2 \sin \theta \cos \theta \pm \sqrt{4 \sin^2 \theta \cos^2 \theta - 4 \cos^2 \theta}}{2 \cos^2 \theta}$$

$$= \frac{-\sin \theta \pm \frac{2 \cos \theta \sqrt{\sin^2 \theta - 1}}{2 \cos^2 \theta}}{\cos \theta}$$

$$= -\tan \theta \pm \frac{\sqrt{-(1 - \sin^2 \theta)}}{\cos \theta}$$

$$= -\tan \theta \pm \frac{i \sqrt{\cos^2 \theta}}{\cos \theta}$$

$$= -\tan \theta \pm i$$

b) i)

$$y = e^{\cos^{-1} x}$$

Domain: $-1 \leq x \leq 1$

Range:

$$0 \leq \cos^{-1} x \leq \pi$$

$$e^0 \leq e^{\cos^{-1} x} \leq e^\pi$$

$$1 \leq y \leq e^\pi$$

ii)

$$x = 0 \quad y = e^{\pi/2}$$

no x -intercept since $y \neq 0$

$$\frac{1}{y} \frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = \frac{-y}{\sqrt{1-x^2}}$$

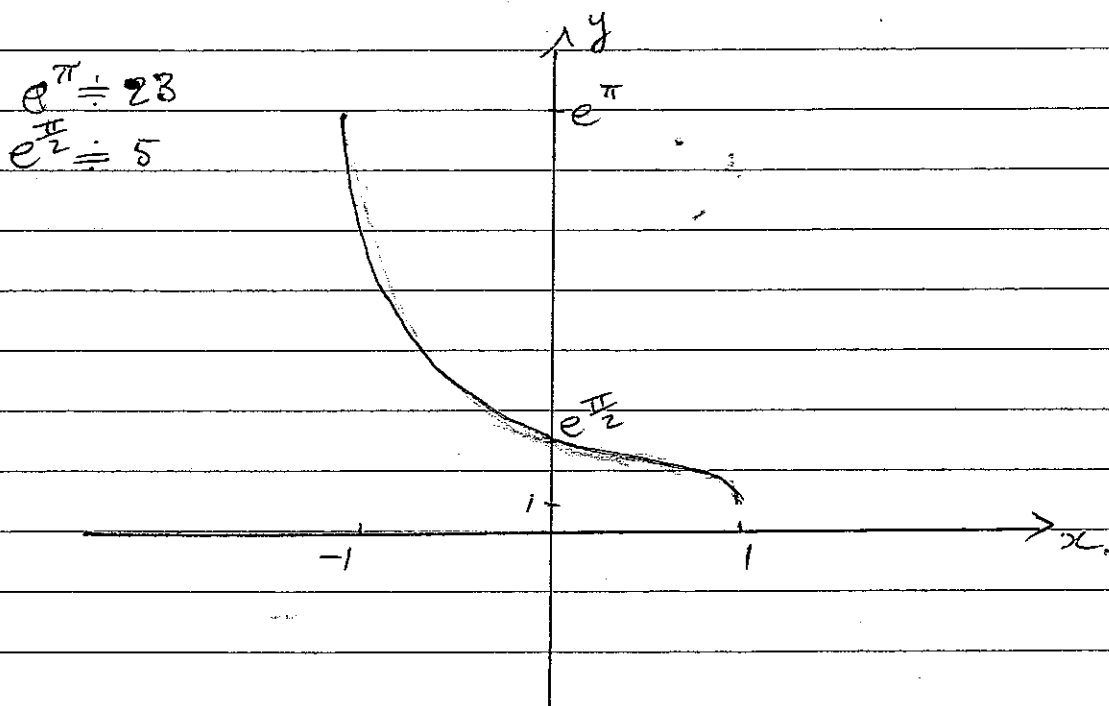
OR If $y = e^{\cos^{-1}x}$

$$\frac{dy}{dx} = e^{\cos^{-1}x} \times \frac{-1}{\sqrt{1-x^2}}$$

$$= \frac{-e^{\cos^{-1}x}}{\sqrt{1-x^2}}$$

Stat pts: $\frac{dy}{dx} \neq 0$ since $y \neq 0$. ✓

Vert tangents: $\frac{dy}{dx} \rightarrow -\infty$ as $x \rightarrow \pm 1$.
since $y > 0$. ✓



$$c) \quad h = a \tan \alpha - \frac{g \sec^2 \alpha}{2V^2} a^2$$

$$= a \tan \alpha - \frac{ga^2}{2V^2} (1 + \tan^2 \alpha) \quad \checkmark$$

$$2V^2 h = 2V^2 a \tan \alpha - ga^2 - ga^2 \tan^2 \alpha$$

$$ga^2 \tan^2 \alpha - 2V^2 a \tan \alpha + 2V^2 h + ga^2 = 0 \quad \checkmark$$

is a quadratic in $\tan \alpha$

has 2 solutions for $\tan \alpha$ if $\Delta > 0$.

i.e. $4V^4 a^2 - 4ga^2(2V^2 h + ga^2) > 0 \quad \checkmark$

$$V^4 - 2gV^2 h - g^2 a^2 > 0.$$

$$V^4 - 2gV^2 h + g^2 h^2 > g^2 a^2 + g^2 h^2$$

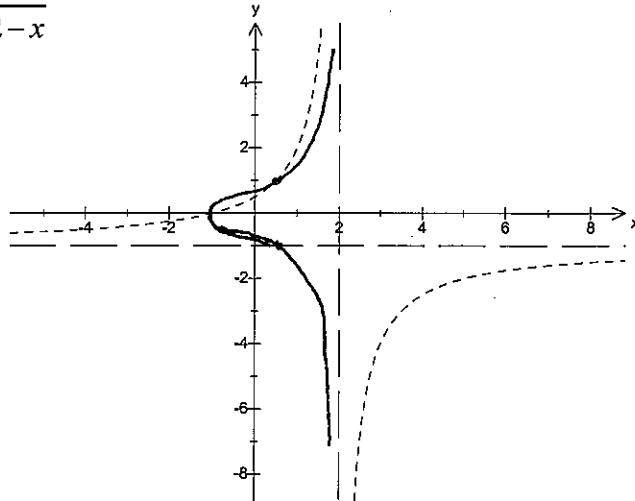
$$(V^2 - gh)^2 > g^2(a^2 + h^2) \quad \checkmark$$

Student Number:

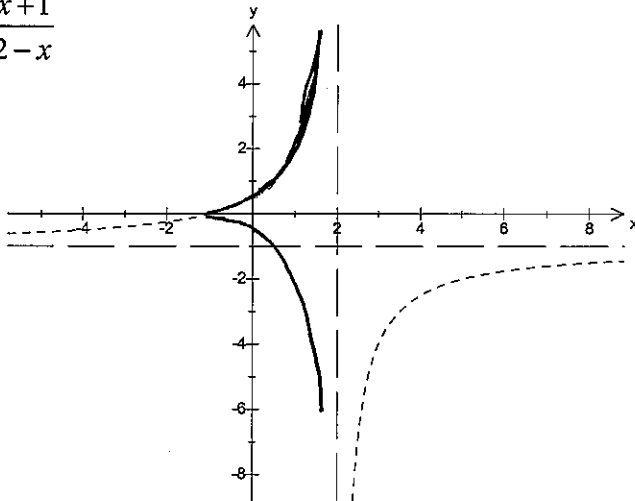
Hand in this page with your answer booklets.

Question 1 (d) Use the graphs of $y = \frac{x+1}{2-x}$ as a template to sketch each of the following:

(i) $y^2 = \frac{x+1}{2-x}$



(ii) $|y| = \frac{x+1}{2-x}$



(iii) $|y| = \frac{|x|+1}{2-|x|}$

